

Grade 9 Student Companion

Ethiopian New Curriculum Integrated Study Book

Mathematics Physics Chemistry Biology English Geography History Civics
ICT Study Skills

Prepared for Knowledge Universe Digital Library

Aligned with the Ministry of Education (MoE) competency-based curriculum framework. This companion summarises core content, worked examples and practice questions for Grade 9 learners.

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Unit	Subject	Focus
1	Mathematics	Number systems, exponents, equations
2	Physics	Measurement, motion, forces, energy
3	Chemistry	Matter, atomic structure, periodic table
4	Biology	Cells, classification, human biology
5	English	Grammar, reading, writing skills
6	Geography	Ethiopia's location, relief and climate
7	History	Ethiopia and the Horn to the 19th century
8	Citizenship	Democracy, rule of law, active citizenship
9	ICT	Computer basics, safety and digital skills
10	Study Skills	Planning, revision and exam technique

How to use this book

- Read one unit per week and complete every Check Your Understanding box.
- Write short notes in your own words copying does not build memory.
- Attempt the practice questions before looking at the answers section of your textbook.
- Use the digital reader's bookmark and progress tracker to keep a steady pace.

Unit 1 Mathematics

1.1 The Real Number System

Numbers are grouped into sets. Natural numbers (1, 2, 3 ...) are used for counting; whole numbers add zero; integers add negatives; rational numbers can be written as a fraction a/b where b is not zero; irrational numbers (such as $\sqrt{2}$ and π) cannot. Together the rational and irrational numbers form the real numbers.

Every real number has a place on the number line, and the order of numbers on that line defines the relations $<$, $=$ and $>$.

Example. Classify $\sqrt{7}$, 0.75, $\sqrt{9}$ and $\sqrt{5}$. Answer: $\sqrt{7}$ is an integer and rational; $0.75 = 3/4$ is rational; $\sqrt{9} = 3$ is a natural number; $\sqrt{5}$ is irrational.

Check Your Understanding

- 1 Simplify: $3/4 + 5/6$
- 2 Is 0.333... rational? Justify your answer.
- 3 Arrange $\sqrt{2}$, $\sqrt{2}$, 1.5, $5/2$ in ascending order.

1.2 Exponents and Radicals

For any non-zero base a and integers m, n : $a^m \cdot a^n = a^{m+n}$, $a^m \div a^n = a^{m-n}$, $(a^m)^n = a^{mn}$, and $a^0 = 1$. A negative exponent means a reciprocal: $a^{-n} = 1/a^n$.

A radical can be written as a fractional exponent: $\sqrt[n]{a} = a^{1/n}$. Scientific notation writes a number as $k \cdot 10^n$ with $1 \leq k < 10$, which is essential in science units.

Example. Simplify $(2^3 \cdot 2^4) \div 2^5 = 2^{3+4-5} = 2^2 = 4$.

Check Your Understanding

- 1 Write 0.00042 in scientific notation.
- 2 Simplify $50 + \sqrt{18}$.
- 3 Evaluate $(3^{-2})^{-1}$.

1.3 Linear Equations and Inequalities

A linear equation in one variable has the form $ax + b = 0$. Solve it by doing the same operation to both sides until the variable stands alone. Inequalities follow the same rules, except that multiplying or dividing by a negative number reverses the inequality sign.

Word problems become equations when you name the unknown, translate each sentence, solve, and finally check the answer against the original situation.

Example. Solve $5x - 7 = 3x + 9$ $2x = 16$ $x = 8$.

Check Your Understanding

- 1 Solve $4(x - 3) = 2x + 6$.
- 2 Solve and graph: $3x + 5 > 11$.
- 3 The perimeter of a rectangle is 46 m and its length is 5 m more than its width. Find the dimensions.

Unit 2 Physics

2.1 Physical Quantities and Measurement

Physics uses seven SI base quantities: length (m), mass (kg), time (s), electric current (A), temperature (K), amount of substance (mol) and luminous intensity (cd). Derived quantities such as speed (m/s) and density (kg/m^3) are combinations of these.

Every measurement carries uncertainty. Record results to the precision of the instrument and always state the unit.

Example. A block of mass 240 g occupies 80 cm³. Density = $m/V = 240/80 = 3 \text{ g/cm}^3 = 3000 \text{ kg/m}^3$.

Check Your Understanding

- 1 Convert 72 km/h into m/s.
- 2 Give the SI unit of force and express it in base units.
- 3 Why is the metre defined using the speed of light?

2.2 Motion in a Straight Line

Distance is a scalar, displacement is a vector. Average speed = $\frac{\text{total distance}}{\text{total time}}$, while velocity includes direction. Acceleration is the rate of change of velocity: $a = \frac{(v - u)}{t}$.

For uniform acceleration: $v = u + at$, $s = ut + \frac{1}{2}at^2$, and $v^2 = u^2 + 2as$. A distance time graph's slope gives speed; a velocity time graph's slope gives acceleration and its area gives displacement.

Example. A car starts from rest and reaches 20 m/s in 8 s. $a = \frac{(20 - 0)}{8} = 2.5 \text{ m/s}^2$; distance = $\frac{1}{2} \times 2.5 \times 8^2 = 80 \text{ m}$.

Check Your Understanding

- 1 A bus travels 150 km in 2.5 h. Find its average speed.
- 2 Sketch a velocity time graph for an object decelerating uniformly to rest.
- 3 Using $v^2 = u^2 + 2as$, find the stopping distance of a 15 m/s cyclist decelerating at 3 m/s².

2.3 Forces, Work and Energy

Newton's laws: (1) a body keeps its state of rest or uniform motion unless a resultant force acts; (2) $F = ma$; (3) forces act in equal and opposite pairs. Weight is the force of gravity, $W = mg$, with $g = 9.8 \text{ m/s}^2$ in Ethiopia's highlands.

Work done $W = F \cdot d$ (when force and displacement are parallel), kinetic energy = $\frac{1}{2}mv^2$, gravitational potential energy = mgh , and power = work / time. Energy is conserved: it changes form but is never destroyed.

Example. Lifting a 20 kg sack 1.5 m: $W = mgh = 20 \times 9.8 \times 1.5 = 294 \text{ J}$. Doing it in 6 s needs a power of 49 W.

Check Your Understanding

- 1 State Newton's second law and give one everyday example.
- 2 Calculate the kinetic energy of a 60 kg runner at 5 m/s.
- 3 Explain the energy changes when a ball is thrown upwards and falls back.

Unit 3 Chemistry

3.1 Matter and Its Properties

Matter exists as solid, liquid, gas and plasma, and changes state when energy is added or removed (melting, boiling, condensation, sublimation). Physical changes alter form only; chemical changes create new substances.

Pure substances are elements or compounds; mixtures can be separated by physical methods such as filtration, evaporation, distillation and chromatography.

Example. Separating salt from sea water uses evaporation a physical change, because the salt is unchanged chemically.

Check Your Understanding

- 1 Classify: milk, oxygen, sodium chloride, air.
- 2 Which method separates alcohol from water? Why?
- 3 Give two observations that show a chemical change has taken place.

3.2 Atomic Structure

An atom has a nucleus of protons (positive) and neutrons (neutral), surrounded by electrons (negative). Atomic number Z = number of protons; mass number A = protons + neutrons. Isotopes are atoms of the same element with different numbers of neutrons.

Electrons occupy energy levels holding at most 2, 8, 8, 18 electrons. The outermost (valence) electrons determine chemical behaviour.

Example. Chlorine-35: $Z = 17$, $A = 35$, so neutrons = $35 - 17 = 18$; electron arrangement 2, 8, 7.

Check Your Understanding

- 1 Draw the electron arrangement of magnesium ($Z = 12$).
- 2 Define isotope and give one example.
- 3 Why is an atom electrically neutral?

3.3 The Periodic Table and Bonding

Elements are arranged by increasing atomic number in periods (rows) and groups (columns). Members of a group share valence electrons and therefore similar properties: Group 1 alkali metals, Group 17 halogens, Group 18 noble gases.

Ionic bonds form when metals transfer electrons to non-metals; covalent bonds form when non-metals share electrons; metallic bonding explains conductivity and malleability.

Example. Na (2,8,1) loses one electron and Cl (2,8,7) gains it, giving Na⁺ and Cl⁻, held together as ionic NaCl.

Check Your Understanding

- 1 Explain the trend in reactivity down Group 1.
- 2 Predict the formula of the compound between magnesium and oxygen.
- 3 Compare ionic and covalent compounds in melting point and conductivity.

Unit 4 Biology

4.1 The Cell

All living things are made of cells. Plant cells have a cell wall, chloroplasts and a large vacuole; animal cells do not. Key organelles: nucleus (control), mitochondria (respiration), ribosomes (protein synthesis), cell membrane (selective transport).

Substances move by diffusion, osmosis (water across a semi-permeable membrane) and active transport, which requires energy.

Example. A wilted plant recovers in fresh water because water enters cells by osmosis, restoring turgor pressure.

Check Your Understanding

- 1 List three structures found only in plant cells.
- 2 Distinguish diffusion from active transport.
- 3 Why do muscle cells contain many mitochondria?

4.2 Classification and Diversity

Living things are grouped into kingdoms and further into phylum, class, order, family, genus and species. The binomial name uses genus and species, e.g. *Homo sapiens*.

Ethiopia is one of the world's biodiversity hotspots, with endemic species such as the Walia ibex, Gelada baboon and Ethiopian wolf. Habitat loss and overgrazing are the main threats.

Example. *Coffea arabica*, native to Ethiopia's south-western forests, illustrates how biodiversity supports the national economy.

Check Your Understanding

- 1 Write the classification hierarchy from kingdom to species.
- 2 Name two endemic Ethiopian animals and their habitats.
- 3 Suggest three conservation actions for a local forest.

4.3 Human Biology and Health

The digestive system breaks food into absorbable nutrients; the circulatory system transports oxygen and nutrients; the respiratory system exchanges gases in the alveoli. A balanced diet supplies carbohydrates, proteins, fats, vitamins, minerals, water and fibre.

Communicable diseases such as malaria, tuberculosis and HIV/AIDS spread through vectors, air or body fluids; prevention depends on hygiene, vaccination, bed nets and health education.

Example. Iron deficiency causes anaemia; leafy vegetables, meat and legumes are good local iron sources.

Check Your Understanding

- 1 Trace the path of a mouthful of injera through the digestive system.
- 2 Explain how malaria is transmitted and prevented.
- 3 Design a one-day balanced menu using locally available foods.

Unit 5 English

5.1 Grammar Essentials

Tense shows time: simple present for habits and facts, present continuous for actions now, simple past for finished actions, present perfect for past actions with present relevance, and future forms with will or going to.

Active voice states who acts (The teacher marked the books); passive voice moves the focus to the action (The books were marked). Subject verb agreement and correct articles (a, an, the) are frequent exam points.

Example. Change to passive: 'Students plant trees every June.' 'Trees are planted by students every June.'

Check Your Understanding

- 1 Rewrite in the present perfect: 'I finish my homework.'
- 2 Correct: 'He don't like mathematics.'
- 3 Turn into reported speech: She said, 'I am reading a novel.'

5.2 Reading and Vocabulary

Use skimming to get the general idea, scanning to find specific facts, and close reading for inference. Guess unknown words from context clues, prefixes and suffixes before reaching for a dictionary.

Keep a personal vocabulary notebook: word, part of speech, meaning, and your own sentence.

Example. In 'The drought decimated the harvest', the surrounding words suggest decimated means severely reduced.

Check Your Understanding

- 1 Skim a newspaper article and write its main idea in one sentence.
- 2 Find five new words in this unit and use each in a sentence.
- 3 What is the difference between a fact and an opinion?

5.3 Writing Skills

A paragraph has a topic sentence, supporting details and a concluding sentence. An essay adds an introduction with a thesis statement, body paragraphs and a conclusion.

Always plan, draft, revise and proofread. Use linking words (however, therefore, in addition, for example) to guide the reader.

Example. Thesis: 'Planting indigenous trees around schools improves both learning and the local climate.'

Check Your Understanding

- 1 Write a 120-word paragraph about your community.
- 2 Draft a formal letter to your school director requesting library books.
- 3 Proofread a classmate's paragraph and list three improvements.

Unit 6 Geography of Ethiopia

6.1 Location, Shape and Size

Ethiopia lies roughly between 3° N and 15° N latitude and 33° E and 48° E longitude, in the Horn of Africa. It is landlocked and borders Eritrea, Djibouti, Somalia, Kenya, South Sudan and Sudan.

Its location gives a tropical position modified by altitude, and a strategic place near the Red Sea trade routes.

Example. Because Ethiopia sits near the equator but at high altitude, Addis Ababa is cool all year – the 'Weyna Dega' climate.

Check Your Understanding

- 1 Name Ethiopia's neighbouring countries.
- 2 How does latitude affect temperature?
- 3 State two advantages and two disadvantages of being landlocked.

6.2 Relief and Drainage

The Great Rift Valley divides the highlands into north-western and south-eastern plateaus. Ras Dashen (about 4,550 m) is the highest point; the Danakil Depression, about 125 m below sea level, is the lowest.

Major rivers include the Abay (Blue Nile), Tekeze, Awash, Wabi Shebelle, Omo and Baro, feeding the Nile, Rift Valley and Shebelle drainage systems.

Example. The Abay supplies the Grand Ethiopian Renaissance Dam, showing how relief and rivers shape development.

Check Your Understanding

- 1 Locate the Rift Valley on a sketch map of Ethiopia.
- 2 Which drainage system does the Awash belong to?
- 3 Explain two effects of rugged relief on transport.

6.3 Climate and Vegetation

Traditional climate zones follow altitude: Bereha (hot desert), Kolla (lowland), Weyna Dega (temperate), Dega (cool highland) and Wurch (afro-alpine). Rainfall comes mainly with the Kiremt rains (June–September) and the shorter Belg rains.

Vegetation ranges from desert scrub and savanna to montane forest and afro-alpine moorland. Deforestation, soil erosion and climate variability are national challenges.

Example. Coffee grows best in the Weyna Dega zone between 1,500 and 2,200 m.

Check Your Understanding

- 1 Match each climate zone to an altitude range.
- 2 Why do the Kiremt rains matter to farmers?
- 3 Suggest three ways to reduce soil erosion on a hillside farm.

Unit 7 History of Ethiopia and the Horn

7.1 Early States and Civilisations

Human ancestors such as Lucy (Dinkinesh) and Ardi place the Horn at the centre of human evolution. Early states included Punt, Da'amat and, from the first century CE, the Aksumite kingdom with its stelae, coinage and Red Sea trade.

Christianity arrived in Aksum in the fourth century CE; Islam reached the Horn in the seventh century through Negash and the coastal trade towns.

Example. Aksumite coins in gold, silver and bronze prove the kingdom's long-distance trade with Rome and India.

Check Your Understanding

- 1 Why is Aksum called a trading civilisation?
- 2 Describe two lasting legacies of Aksum.
- 3 How did religion shape early Ethiopian states?

7.2 Medieval States and Interaction

The Zagwe dynasty (about 1137–1270) built the rock-hewn churches of Lalibela. The 'restored' Solomonic dynasty followed, coexisting and competing with Muslim sultanates such as Ifat and Adal, and with the Oromo population movements of the sixteenth century.

Trade in gold, ivory, coffee and salt bars (amole) linked highland and lowland peoples.

Example. The Lalibela churches, carved from single blocks of rock, are a UNESCO World Heritage site.

Check Your Understanding

- 1 Who were the Zagwe and what are they remembered for?
- 2 Explain two causes of conflict between the Christian kingdom and Adal.
- 3 How did the Gadaa system organise Oromo society?

7.3 Towards the Modern State

The Zemene Mesafint (Era of Princes, about 1769–1855) was a period of regional lords. Reunification began with Tewodros II, continued under Yohannes IV and was consolidated by Menelik II, whose victory at Adwa in 1896 defeated Italian colonial ambitions.

Adwa became a symbol of African independence and inspired pan-African movements.

Example. At Adwa on 1 March 1896 Ethiopian forces under Menelik II and Empress Taytu defeated the Italian army.

Check Your Understanding

- 1 List the main achievements of Tewodros II.
- 2 Why is the Battle of Adwa historically important?
- 3 Describe two modernisation measures taken by Menelik II.

Unit 8 Citizenship and Moral Education

8.1 Democracy and the Constitution

Democracy rests on popular sovereignty, free and fair elections, separation of powers and respect for human rights. The Constitution is the supreme law; any law that contradicts it is invalid.

Citizens have rights (life, equality, expression, education) and duties (obeying the law, paying taxes, protecting public property, respecting others' rights).

Example. Voting in an election is both a right and a responsibility of every eligible citizen.

Check Your Understanding

- 1 Distinguish a right from a duty with examples.
- 2 Why is the rule of law important?
- 3 Name the three branches of government and one function of each.

8.2 Active Citizenship and Ethics

Active citizens volunteer, protect the environment, resist corruption and promote tolerance among Ethiopia's diverse nations, nationalities and peoples. Ethical decision-making asks: is it legal, is it fair, and how will it affect others?

Conflict is best resolved by dialogue, negotiation and mediation rather than violence.

Example. A school anti-litter club that organises weekly clean-ups is practical active citizenship.

Check Your Understanding

- 1 Plan a small community service project for your class.
- 2 Give three effects of corruption on development.
- 3 Describe a peaceful way to settle a dispute between two students.

Unit 9 Information and Communication Technology

9.1 Computer Fundamentals

A computer system has hardware (input, processing, storage, output devices) and software (system software such as the operating system, and application software). Data becomes information when it is processed and given meaning.

Storage is measured in bytes: 1 KB = 1024 B, 1 MB = 1024 KB, 1 GB = 1024 MB.

Example. A keyboard is an input device; the CPU processes data; a monitor outputs it; a hard disk stores it.

Check Your Understanding

- 1 Classify: mouse, printer, RAM, operating system.
- 2 Convert 2 GB into megabytes.
- 3 Explain the difference between data and information.

9.2 Digital Skills and Online Safety

Word processors, spreadsheets and presentation tools support learning. Use strong, unique passwords, avoid sharing personal details, verify sources before believing them, and report cyberbullying.

Respect copyright: cite sources and avoid plagiarism when using online material for assignments.

Example. A strong password mixes upper and lower case letters, digits and symbols and is at least 12 characters long.

Check Your Understanding

- 1 List four rules for safe internet use.
- 2 How can you check whether an online article is reliable?
- 3 Create a spreadsheet plan for tracking your weekly study hours.

Unit 10 Study Skills and Exam Technique

10.1 Planning Your Learning

Set SMART goals – specific, measurable, achievable, relevant and time-bound. Build a weekly timetable that fixes school, study, chores and rest, and protect a daily one-hour focused study block.

Study in 25–30 minute blocks with 5-minute breaks; the brain retains far more from short focused sessions than from long tired ones.

Example. Goal: 'Solve 20 linear-equation problems and score 8/10 on a self-test by Friday.'

Check Your Understanding

- 1 Write three SMART goals for this term.
- 2 Draw your weekly timetable.
- 3 Which time of day do you concentrate best, and why?

10.2 Revision and Assessment

Use active recall (close the book and write what you remember), spaced repetition (review after 1 day, 3 days, 1 week), summary cards and past papers. Teaching a topic to a classmate is one of the strongest revision methods.

In an exam, read all instructions, allocate time per mark, start with the questions you know, show your working, and leave time to check.

Example. For a 2-hour, 100-mark paper, roughly one minute per mark leaves about 20 minutes for checking.

Check Your Understanding

- 1 Make five recall cards for Unit 3.
- 2 Plan a revision schedule for four subjects in two weeks.
- 3 List three exam mistakes you will avoid this year.

Glossary of Key Terms

Term	Meaning
Rational number	A number that can be written as a fraction a/b with $b \neq 0$.
Acceleration	Rate of change of velocity, measured in m/s^2 .
Isotope	Atoms of one element with different numbers of neutrons.
Osmosis	Movement of water across a semi-permeable membrane.
Endemic species	A species found naturally in one region only.
Thesis statement	One sentence stating the main argument of an essay.
Rift Valley	A long depression formed where the Earth's crust pulls apart.
Zemene Mesafint	The Era of Princes, a period of regional rule before reunification.
Rule of law	The principle that everyone, including leaders, is subject to the law.
Active recall	Revising by retrieving information from memory without notes.

Study Charter

I will study a little every day, ask questions when I do not understand, help a classmate each week, and respect my school, my community and my environment.